

Final Project Report

EE 3300
Temperature Sensor

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Lab Section A

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Objective

Our objective for this final project consisted of designing a temperature sensor that provided an analog output that varied linearly with PTAT (Proportional to Absolute Temperature). We were also required to operate over the range of 0°C to 50°C with the output being 0V at 0°C and 1V at 50°C. Finally, we were required to implement this design by using, at minimum, diodes and an operational amplifier.

Overview

To implement our design, we broke up the circuit into two stages. Stage one consisted of two diodes that acted as the temperature sensors, a current mirror to ensure that the diodes had the same current, and a resistor that served as the PTAT variable. Stage two consisted of a differential operational amplifier and a resistor network. Lastly, our layout consisted of a few different components including PMOS's, resistors, the differential op amp, and diodes.

Stage 1

Current Mirror

In a broad sense, we had to use a current mirror because we needed the diodes to have equal current. This makes it easier to do calculations because current is then eliminated as a variable.

To make sure this happened, we made $\frac{W_1}{L_1} = \frac{W_2}{L_2}$ so that the current would be equal on both sides of the mirror. If the currents are not the same, the diode equation results in a current ratio rather than an area ratio. The area ratio is much easier to control.

Diodes

Since the diodes have different areas, this creates a difference in current densities. Therefore, it creates a voltage difference, which is what we want. If given two diode equations, we can set up an equation that models the difference in voltages. This result will then give us an indication as to how temperature varies linearly with voltage depending on the areas. The work for this derivation can be seen below.

$$I_D = I_S e^{\frac{V_D}{V_t}} \rightarrow V_D = V_t \ln\left(\frac{I_D}{I_S}\right)$$

$$V_{D1} = V_t \ln\left(\frac{I_D}{I_{S1}}\right) \quad V_{D2} = V_t \ln\left(\frac{I_D}{I_{S2}}\right)$$

$$\begin{aligned} \Delta V_D &= V_{D1} - V_{D2} = V_t \left(\ln\left(\frac{I_D}{I_{S1}}\right) - \ln\left(\frac{I_D}{I_{S2}}\right) \right) \\ &= V_t \left(\ln\left(\frac{I_{S2}}{I_{S1}}\right) \right) \end{aligned}$$

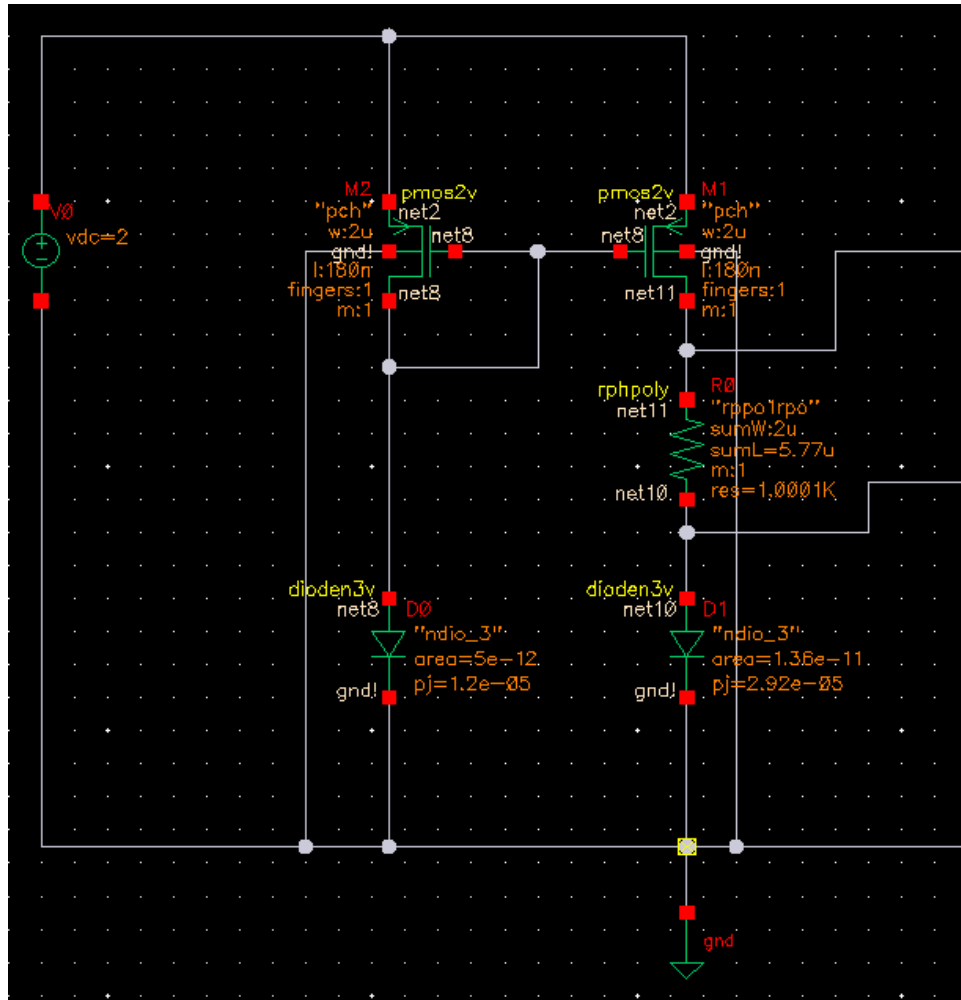
$$I_S = J_S A_E \rightarrow \Delta V_D = V_t \ln\left(\frac{A_{E2}}{A_{E1}}\right)$$

$$\frac{A_{E2}}{A_{E1}} = 2.72 \rightarrow \ln(2.72) = 1$$

$$\Delta V_D \propto T \rightarrow PTAT$$

As the reader can see, this derivation proves that temperature varies linearly with the voltage change thanks to the sizing of the diodes. For our purposes, and so that the numbers look cleaner, we made the area of D2 $13.6\mu^2$ and D1 $5\mu^2$. This makes it so that the ratio stays at about 2.72.

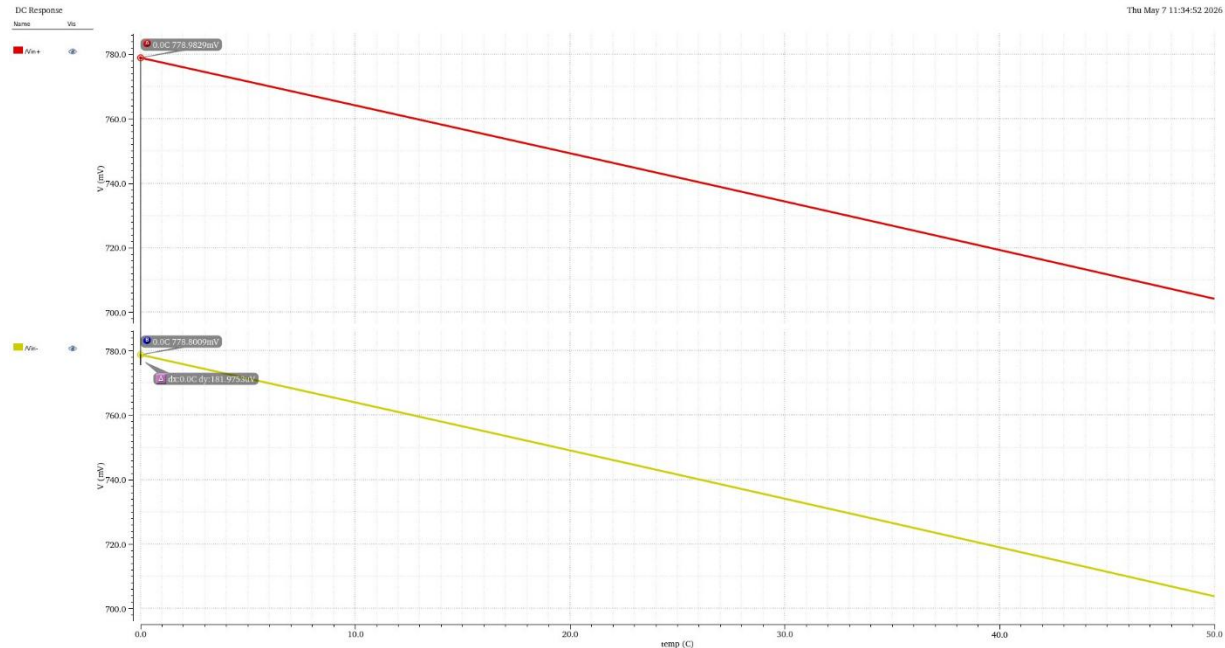
Our final circuit for stage 1 of the design can be seen below. We chose R_{PTAT} to be $1\text{ k}\Omega$ because the sizing of this resistor actually did not matter too much. This is because in our design, instead of relying on a large voltage difference across the resistor, the gain from the differential amplifier gives the range of voltage needed.



Stage 2

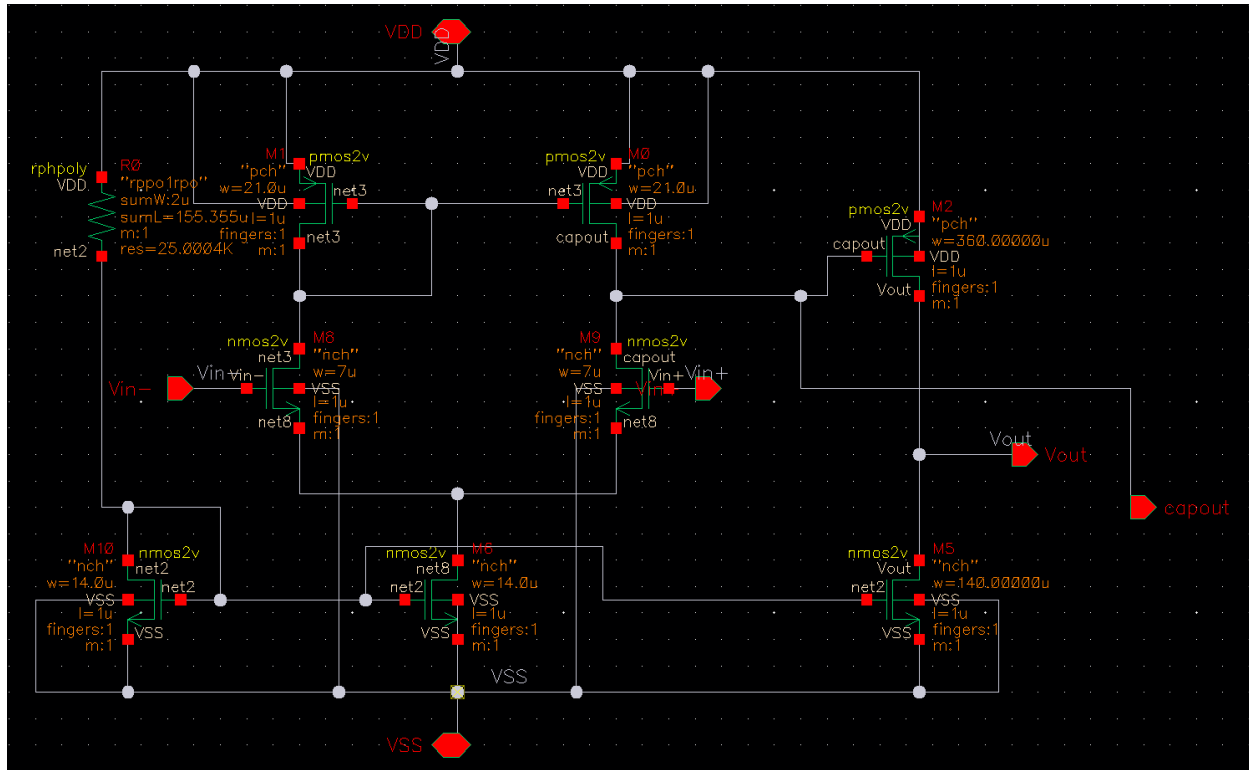
Differential Operational Amplifier

This circuit needed an operational amplifier due to the fact that the output voltage difference from R_{PTAT} was extremely small. Also, this operational amplifier needed to have a differential aspect to it because there are two input signals that are coming into it. This is an issue, because we need one output voltage, not two. Therefore, we used a differential operational amplifier because it takes the difference between the two input terminals and outputs the difference. This can be further demonstrated in the graph below. It may be a bit hard to see, but the graph is showing the positive input terminal voltage against the negative input terminal voltage into the operational amplifier. The difference is very small, only about $182 \mu\text{V}$. It's a very small output voltage, but now there is only one output signal.



The next task to fulfill with this differential operational amplifier was to actually design it. We referred to the design given within the EE330 website to aid in this task. The design consisted of eight transistors with varying width and length ratios. Also, it consisted of a current source that we had to design with a resistor because just placing an ideal current source into the design was out of the scope of the project. This design can be seen below, along with explanations for the sizing of each transistor.

To start off the design of the differential operational amplifier, the first step is to pick a bias current value. The initial design was made with a current source which was later replaced with a resistor. The chosen value for the bias current was $120\text{ }\mu\text{A}$. The NMOS sizes are then made in a way that divides the current leading to them all being in the saturation region. When testing, the saturation region corresponds to region 2. For sizing all the MOS's have the minimum length of $1\text{ }\mu\text{m}$. The bottom NMOS's form a current mirror, with the first two being the same size so that they have the same current. Under the second branch that feeds into the second current mirror NMOS, M6, due to the reference sheet provided on the EE330 website it is known that the inputs of v_{in-} and v_{in+} should feed into transistors of the same size, which is why M8 and M9 are the same size. Looking at the PMOSs' M1 and M0 they also need to be the same sized but due to hole mobility being slower than electron mobility by about three times, the PMOS need to be three times bigger in width. Finally, M2, which is the second stage of the operation amplifier, is made large in order to support a large gain as well as increase stability, but because M2 is so large, M5, the last part of the current mirror needs to also be large in order to provide enough current so M2 is in saturation.



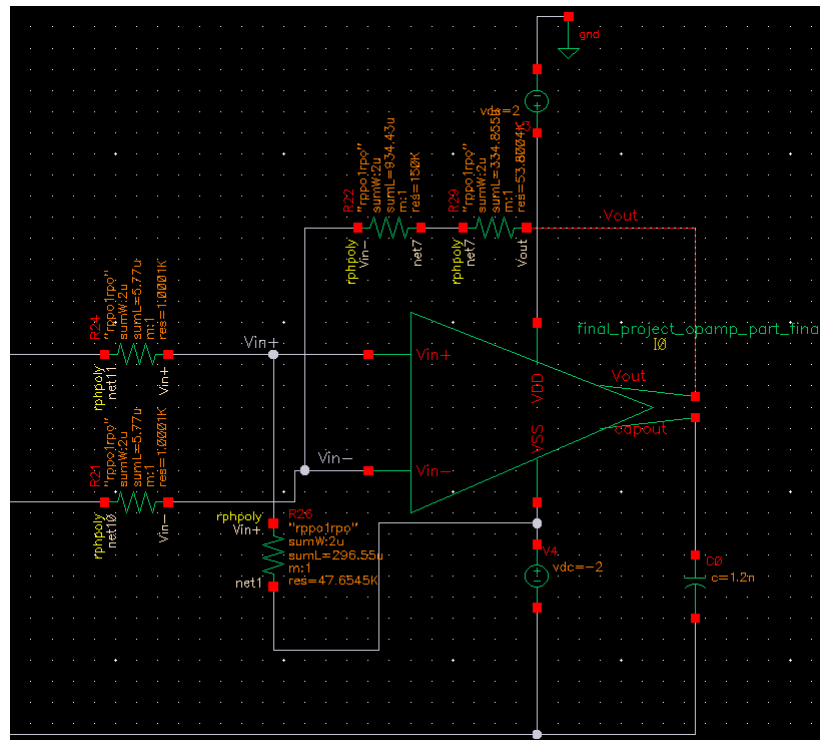
Resistor Network

Our design in stage 2 consisted of a resistor network. On the input to the positive voltage port of the amplifier, there is a voltage divider. The purpose of this voltage divider is to create an offset and scale the output voltage from the starting point at V_{REF} , which is our VSS value of -2V, and bring it to zero. This task required a great deal of trial and error. We decided to keep the resistor that was not connected to ground at 1 k Ω so that we only had to adjust the resistor connected to V_{REF} . In the end, the resistor connected to V_{REF} had a value of 47.6545 k Ω . We observed that adjusting the scale was VERY tedious. If we incremented the value by 0.01, the scale could overshoot or undershoot drastically compared to how much we thought it would change. To figure out the initial value of the resistor we first measure the different voltages flowing into the two terminals, because if the voltages are the same then the output of the operational amplifier will be 0. We found that for V_{in+} to match V_{in-} a voltage divider of $\frac{1}{48}$ was needed. Since the initial resistor is 1k Ω that meant the resistor connected to V_{ref} should be 47k Ω , minor adjustments to the value after that reduced the error of the circuit.

Additionally, our design in stage 2 had an inverting amplifier feedback resistor network connected to the negative voltage port of the amplifier. This is because the differential operational amplifier also acts as an inverting amplifier. These resistors were there to amplify the output voltage coming from the PTAT voltage difference. As said before, the difference was very small, so our gain had to be large. Again, we left the resistor that was not in the feedback network at 1 k Ω so that the feedback resistor was the only one we had to adjust. This task also

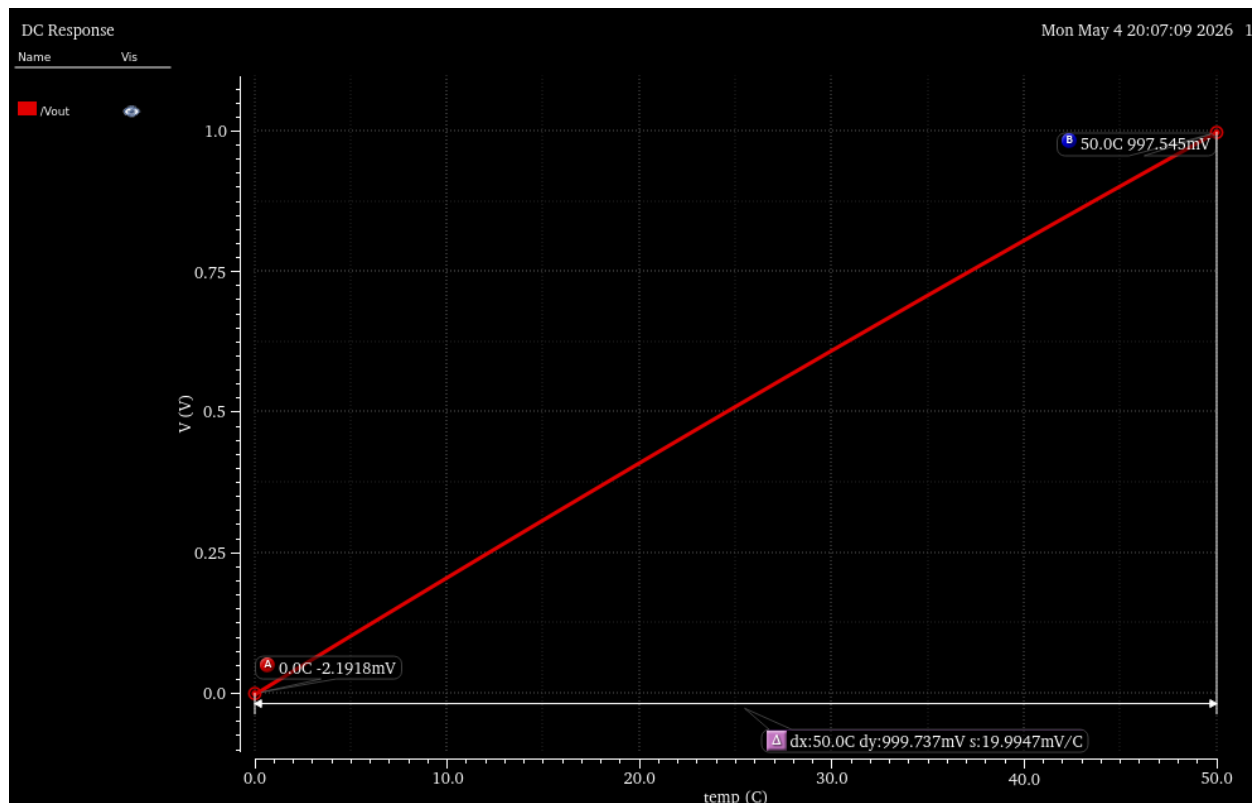
required a ton of trial and error. Before being multiplied by the gain the output of the differential amplifier was around 5.06 mV. We needed to get a gain of 200 in order to reach the values we needed. In the end, the feedback resistor ended up being 203.8004 k Ω . This value was also very tedious. One small adjustment would overshoot or undershoot our gain compared to what we wanted it to be. Overall, our gain, determined by the feedback resistor and the resistor connected to the negative voltage port, turned out to be -203.8004.

Our final circuit for stage 2 can be seen below. The capacitor connected to the operational amplifier is 1.2nF because it increases the stability of the operational amplifier as it further separates the values of the poles of the transfer function. This leads to one of the poles being at a very low frequency and the other pole being at a very high frequency.



Final Circuit

Our final circuit can be seen below. The first stage is seen on the left while the second stage is seen on the right.



Our required slope for this project was 0.02, since the graph rises 1 and runs 50. Our measured slope was 0.0199947. This means that our output was 99.99735% accurate to our goal.

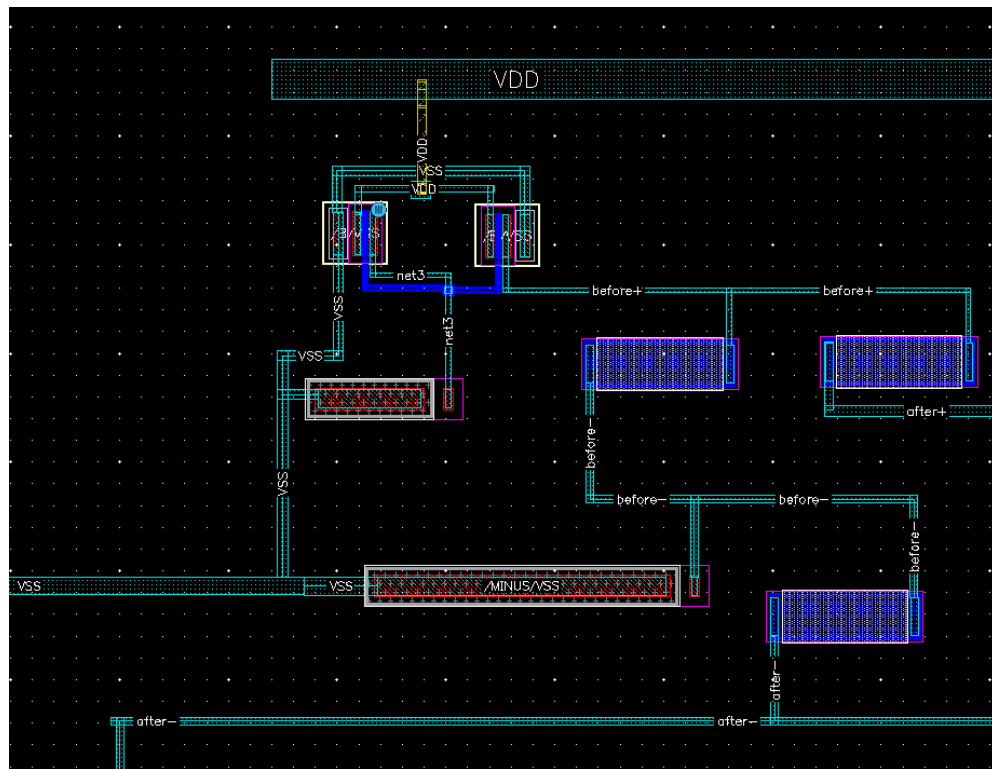
Layout

Like our final circuit design, we broke up our layout into two stages. This layout was a little more challenging than our ones in the past with the previous labs. This is because there were several MOSFETs that had different sizes and the resistors had different sizes. This made the layout of the device much larger which made it much more time consuming to complete the layout. Also, we were having major issues with our LVS testing results. This will be discussed in one of the following sections. The materials used to connect the ports to their specific components were mostly Metal 1, with a bit of Metal 2 involved when the same metal could cross paths at a place where we didn't want it to. Finally, Poly 1 was used when connecting the gates of the transistors to their respective locations.

Stage 1

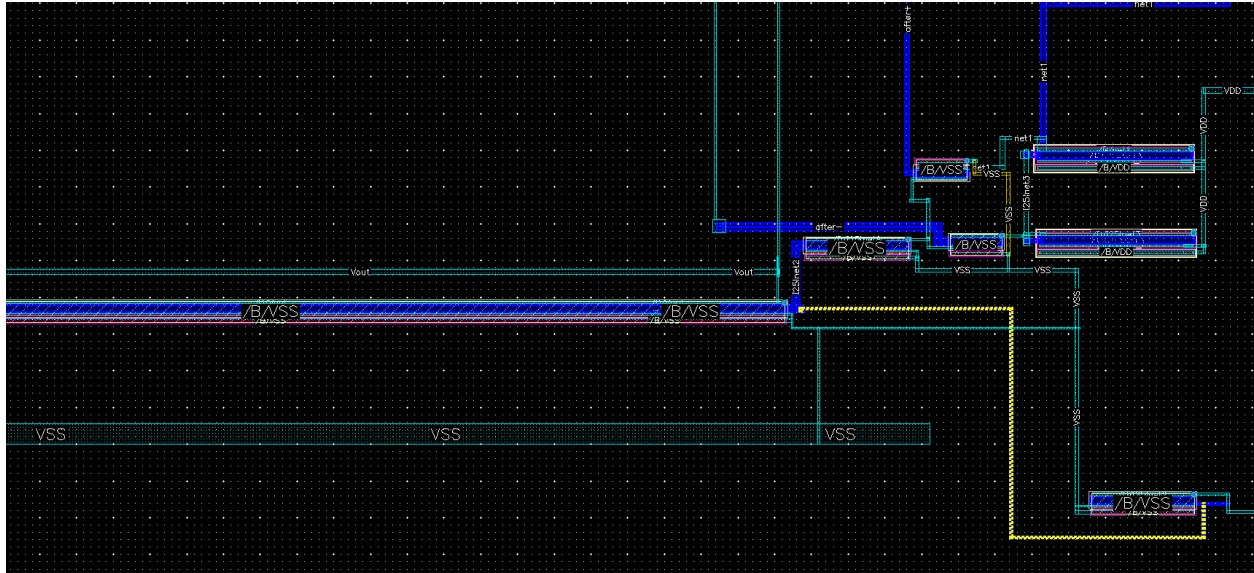
The first stage of this layout correlates directly with the first stage of our designed circuit. It consists of the two PMOS transistors along with the two diodes and R_{PTAT} . Additionally, we put in the two resistors that input to the positive and negative terminals of the operational amplifier. These three resistors were all 1 k Ω , so they are the same size on the layout. The transistors have

the same width and length ratios, so they are also the same size in the layout. Lastly, since the diodes have different areas, they are different sizes on the layout, which can be seen on the image below.



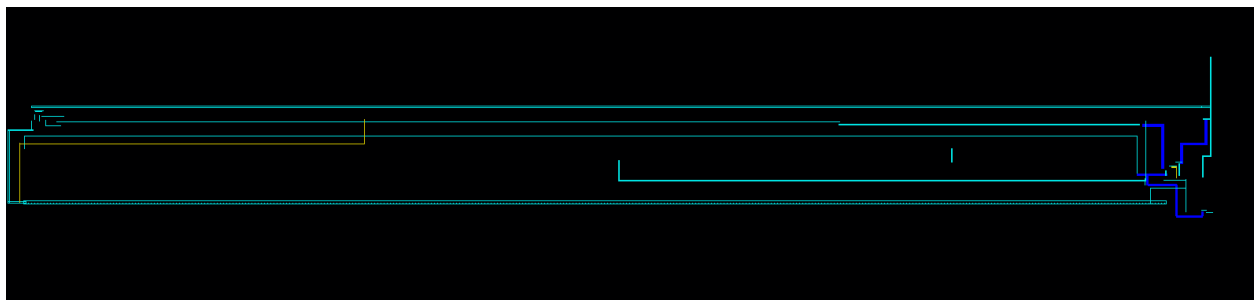
Stage 2

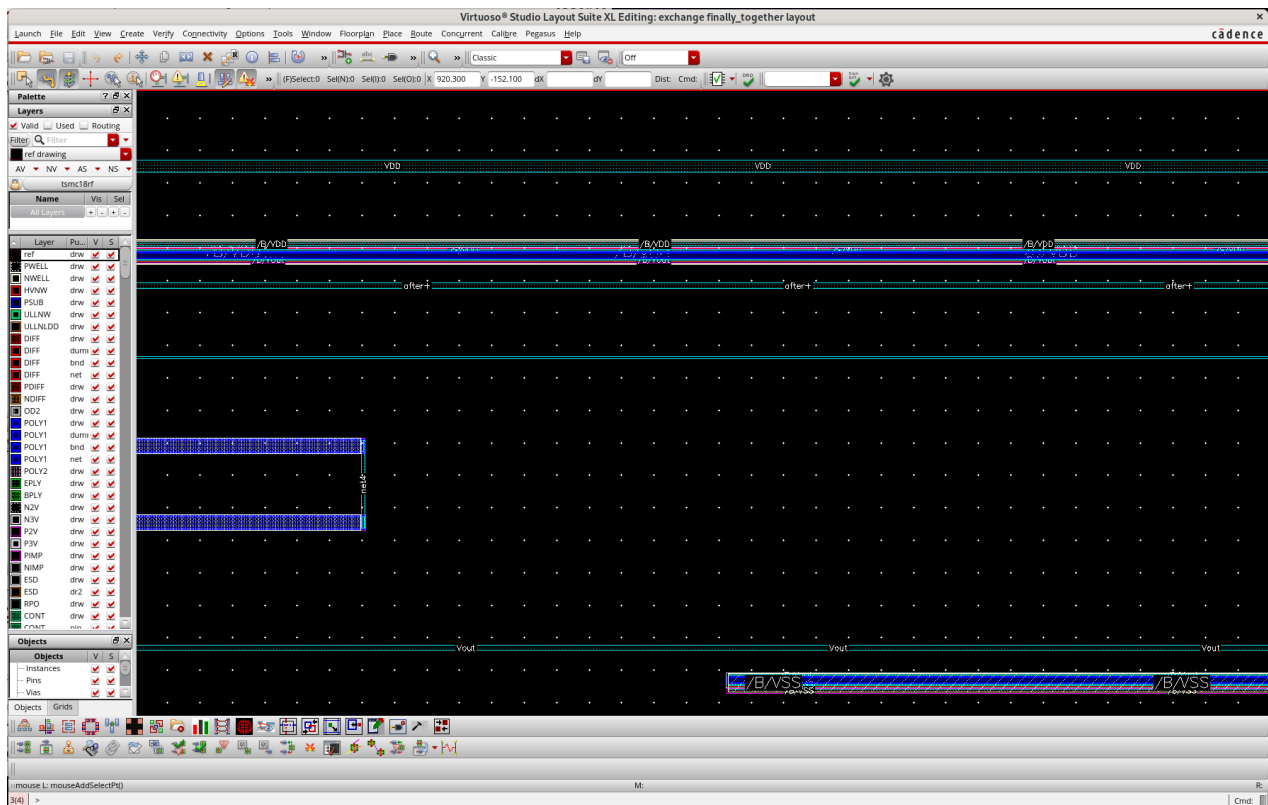
Again, the second stage of the layout correlated directly with the second stage of our circuit design. It consisted of the operational amplifier transistors along with the resistors that made up the voltage divider and feedback network of the amplifier. Unfortunately, since our resistors are so big and one of the MOSFETs is so large, we couldn't fit everything into a single screenshot. The image below shows seven of the eight transistors, but none of the resistors. These will be seen in the final layout screenshot.



Final Circuit

Our final layout can be seen below. As the reader can see, it is a bit hard to make out exactly what is going on. This is because our resistors' values were just so large. For reference, the small collection of devices on the far left-hand side of the image is the first stage of the circuit. Do you see the small, vertical strip of Metal 1 that stands all by itself in the middle right section of the screenshot? This is a connection that connects the 150 k Ω resistor to the 53.8004 k Ω resistor in series. The bigger resistors connect from the area where the vertical strip of Metal 2 is on the far left side of the screenshot, all the way over to that Metal 1 strip that was mentioned earlier. The screenshot is just so far out that it cannot detect these components. Additionally, the eighth transistor that was not included in the second stage screenshot is in this one. You cannot see it, but it is located above one of the long, horizontal Metal 1 strips. To prove that what we are saying is true, and that our layout actually worked, we included an additional screenshot that included the resistors and the final MOS. Also, it includes the vertical Metal 1 strip that has been discussed. Lastly, it includes the title of the layout schematic at the top to show that it is our design.

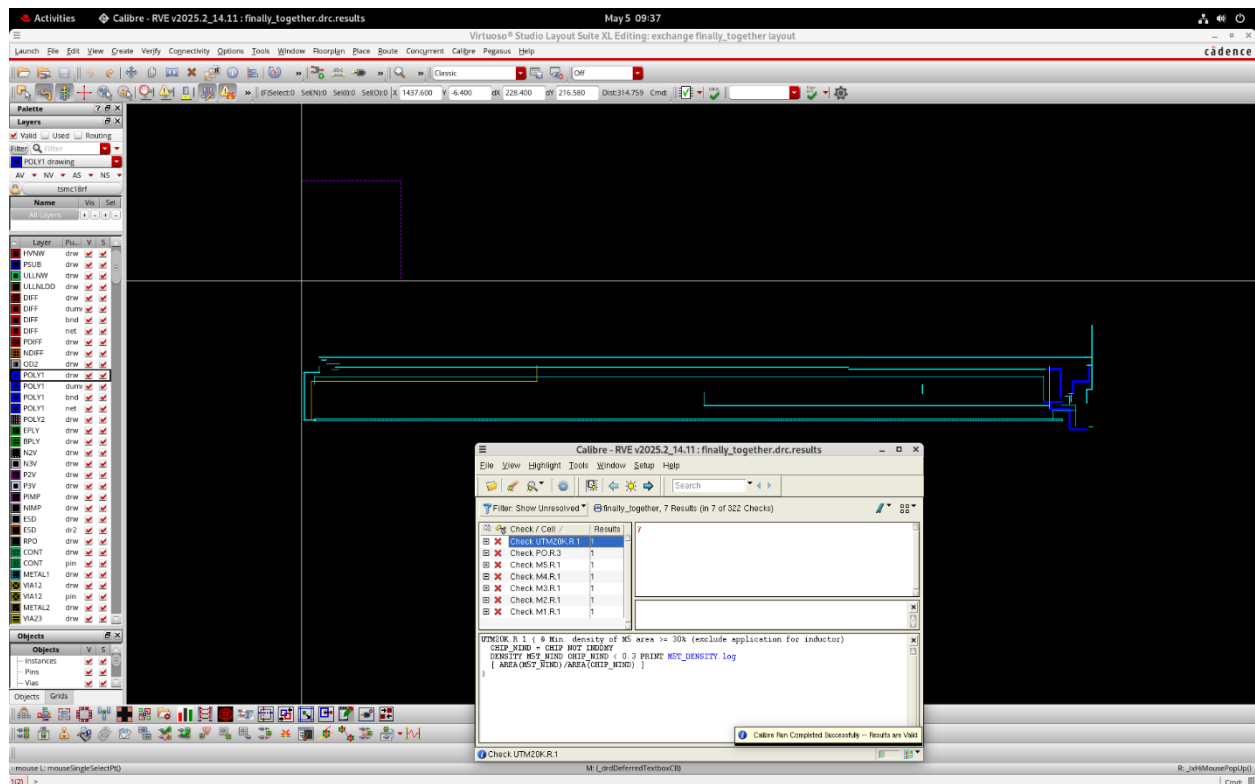




DRC and LVS Test Results

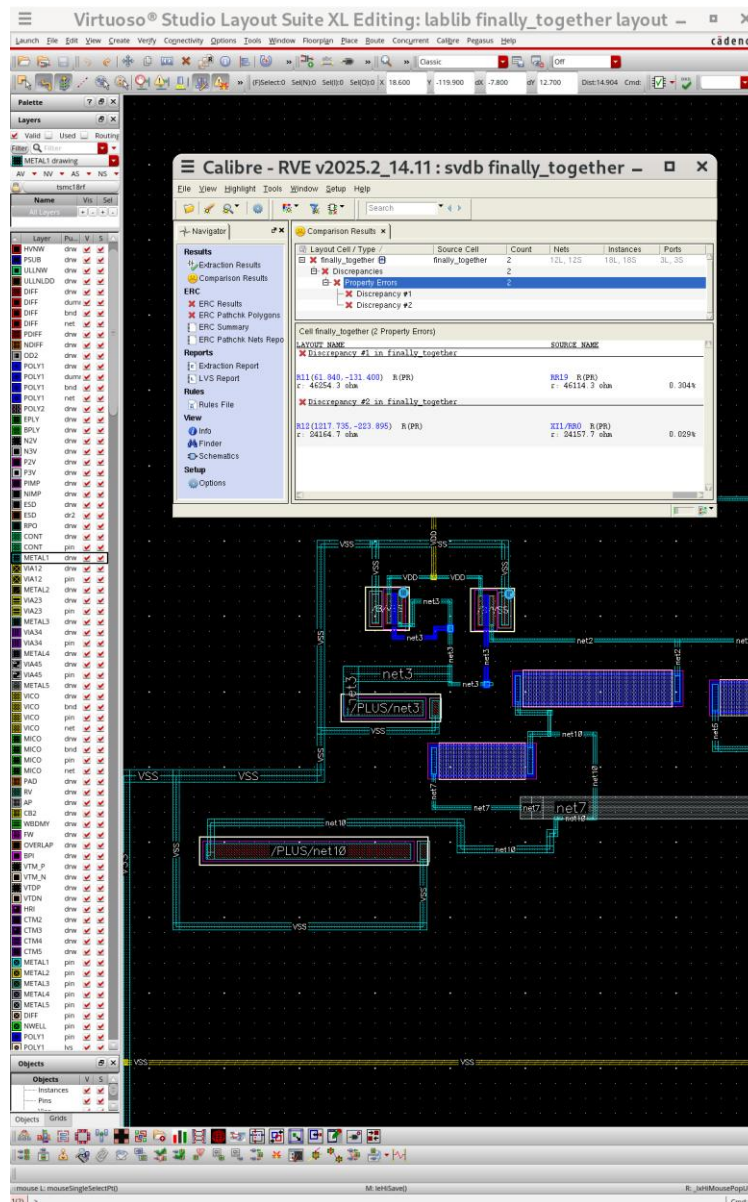
DRC Testing

This section goes over our testing for the layout and schematic of our circuit. The DRC testing went very well. The only issues we had were a few instances where a Metal 1 strip was too close to another device or a strip was too thin. These were easy fixes and our DRC results can be seen below. The only errors that appear are the ones that have been valid throughout the entire semester corresponding to the density of certain materials.



LVS Testing

On the contrary, our LVS results were a much more challenging issue. When we initially ran the test, we encountered multiple errors consisting of property errors, instance errors, and net errors. The property errors were due to the software not allowing the resistor values to be certain values. Therefore, we had to change the resistor values so that they were within 1% of what the expected values were. When it came to the instance errors, we had a very tough time figuring out what was going on. As it turns out, the software did not recognize the “dioden3v” instance within the circuit. We had to figure this out through multiple hours of trial and error. Our solution was to switch out dioden3v with diodep3v. We first discovered this might be the issue when we tried to replace the diodes with diode connected BJTs and the instance errors disappeared. We then experimented with implementing different types of diodes as the BJTs didn’t allow us to customize the emitter size as accurately as we wanted. In doing this we found that the p-doped diodes work but all instances of n-doped diodes created instance errors. The p-doped diode implementation swapped the dope regions of the diodes, but we still managed to get the circuit to behave as before. Finally, the net error was caused because the software did not recognize our diodes. After we switched them out, the errors went away. Below is our LVS test. As the reader can see, we still have two property errors. This was due to the resistors for some reason, and we were instructed to just adjust the values so that the error percentage was under 1%, and we did just that.



this to work properly we would've needed to make a voltage follower that would block the currents so that it wouldn't draw extra current from the diodes. This was a lot more complicated than just taking the voltage difference across the PTAT resistor, so we decided to switch our design. It also would require two additional operational amplifiers to be implemented. If we were to redo this project in the future, we would choose not to use diodes, and instead use diode connected BJTs to avoid LVS issues when doing the layout.